

Detecting Sub-threshold Eavesdropping in BB84 Quantum Key Distribution with Machine Learning

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Research Scholars Day 2026

Abstract

The security of the BB84 quantum key distribution (QKD) protocol is conventionally certified by a single statistic: if the quantum bit error rate (QBER) exceeds the $\sim 11\%$ threshold, the key is discarded. We show that this operational test is blind to a class of *sub-threshold* eavesdropping attacks that hold the QBER below the alarm level while leaving distinct statistical fingerprints elsewhere in the channel. Using a photon-number/decoy-state BB84 simulator — validated against closed-form QKD theory — we model two such attacks: a basis-biased intercept–resend attack (per-basis QBER asymmetry) and a *gain-matched* photon-number-splitting (PNS) attack (decoy-gain collapse at normal QBER and normal signal gain). The operational QBER test detects 0.8% of these attacks; a random forest on a richer feature set detects 87% (AUC 0.96). The gain-matched PNS attack is *invisible* to QBER (AUC 0.51, i.e. random) yet detected perfectly (AUC 1.00) through the decoy/signal gain ratio alone, exactly as decoy-state theory predicts. Finally, a quantum-kernel SVM with a naive encoding fails (AUC 0.58) due to exponential kernel concentration, but becomes competitive with classical baselines once the encoding is tuned (AUC 0.95) — competitive, but with no quantum advantage on this low-dimensional task.

1 Introduction

Quantum key distribution lets two parties, Alice and Bob, share a secret key whose security rests on physics rather than computational hardness. In BB84 [1], Alice sends qubits in one of two conjugate bases and Bob measures in a randomly chosen basis; after public basis reconciliation (*sifting*), the residual disagreement rate — the QBER — bounds an eavesdropper’s information. The textbook security check is a threshold: if $\text{QBER} > \sim 11\%$, eavesdropping is assumed and the key is aborted [2].

This single-number test only sees the *average* error rate. A sophisticated eavesdropper can deliberately keep the QBER below the threshold while still disturbing the channel in ways that are statistically detectable in other observables. We ask: **can machine learning detect eavesdropping that the QBER threshold misses?** We answer affirmatively, identify which physical observables carry the signal, and report an honest assessment of whether a quantum kernel helps.

2 Background

The QBER threshold. For symmetric intercept–resend each intercepted qubit incurs a 25% error, so a full attack drives QBER to $\sim 25\%$, far above the bound. The threshold test is effective against such overt attacks but assumes the eavesdropper operates near full strength.

Basis-biased intercept–resend. If Eve intercepts only a fraction of pulses and always measures in a fixed basis (Z), her errors concentrate in the conjugate (X) basis. The *total* QBER stays below 11%, but the per-basis rates become asymmetric, $\text{QBER}_X \gg \text{QBER}_Z$ — a signature the averaged test discards.

PNS and decoy states. Real sources emit weak coherent pulses with a Poissonian photon number; some pulses contain ≥ 2 photons. In a PNS attack Eve blocks single-photon pulses and splits one photon from each multi-photon pulse, forwarding the rest. Crucially, a competent Eve forwards through a *lossless* line and throttles her forwarding so that Bob’s *signal gain matches* the expected lossy value — masking her interference. She adds no error (she does not measure–resend), so QBER is unchanged too. Because she cannot distinguish signal from decoy pulses, the same throttling is applied to low-intensity *decoy* pulses, whose multi-photon fraction is far smaller; their gain therefore collapses [3]. The decoy/signal gain ratio is the *only* remaining signature — which is precisely why decoy-state QKD was invented.

3 Methods

3.1 Simulator

We extend a Qiskit-Aer BB84 simulator with a photon-number/decoy-state channel. Each pulse is randomly signal ($\mu_s = 0.5$) or decoy ($\mu_d = 0.1$), each with probability tuned by `decoy_prob=0.5`; its photon number is $k \sim \text{Poisson}(\mu)$ and it is detected with probability $1 - (1 - \eta)^k$ at transmittance $\eta = 0.15$ (a high-loss channel, the regime where PNS is a genuine threat). Detected qubits are prepared, sent through a depolarizing channel (3% gate error), and measured on a real single-qubit circuit. The simulator returns total and per-basis QBER, the sift ratio, and the signal/decoy gains. Bob’s measurement sampler is seeded, so all results are reproducible.

For the gain-matched PNS attack, Eve forwards multi-photon pulses with probability $q = (1 - e^{-\mu_s \eta}) / P(k \geq 2 \mid \mu_s)$, which makes the observed signal gain equal the legitimate value $1 - e^{-\mu_s \eta}$.

3.2 Theory validation

Before any learning, we checked the simulator against closed-form QKD expressions (Table 1); all observables agree to within sampling error.

Table 1: Simulator vs. theory ($\mu_s=0.5$, $\mu_d=0.1$, $\eta=0.15$).

Observable	Simulated	Theory
Secure signal gain $1 - e^{-\mu_s \eta}$	0.223	0.221
Secure decoy/signal ratio	0.215	0.221
PNS signal gain $P(k \geq 2 \mid \mu_s)$	0.089	0.090
PNS decoy/signal ratio	0.055	0.052
Secure QBER (3% depolarizing)	0.020	0.02–0.03
Biased-IR asymmetry ($\approx \text{strength}/2$)	0.115	0.100

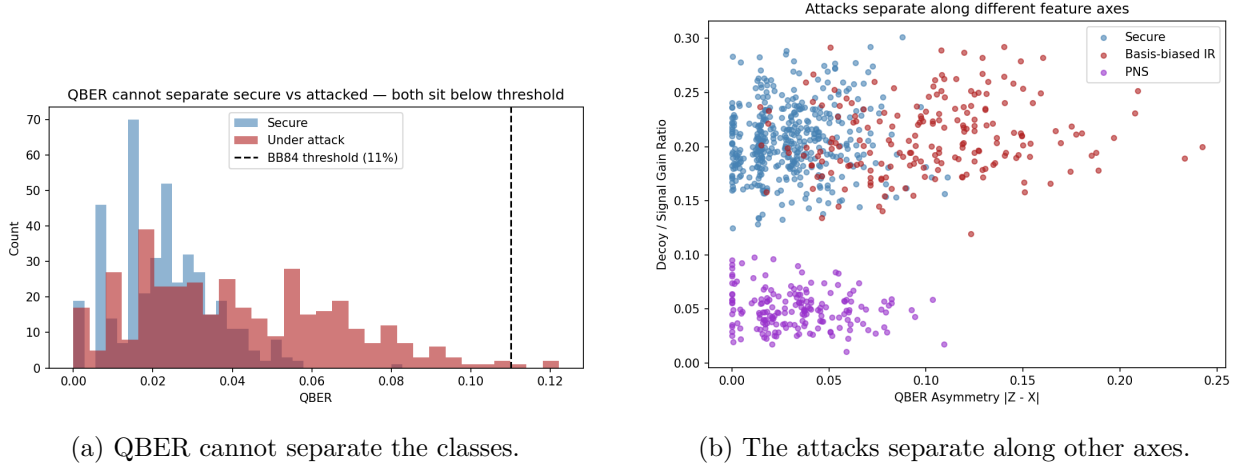


Figure 1: Left: secure (blue) and attacked (red) sessions have overlapping, sub-threshold QBER. Right: basis-biased and PNS attacks are cleanly separable in the (asymmetry, decoy-ratio) plane.

3.3 Attacks, features, models

The basis-biased attack intercepts a fraction $p \in [0.08, 0.22]$ of pulses (chosen to keep total QBER below 11%); PNS is gain-matched as above. Each session yields five features: total QBER, per-basis QBER asymmetry $|\text{QBER}_Z - \text{QBER}_X|$, signal gain, decoy/signal gain ratio, and sift ratio. We train logistic regression, a random forest, and an SVM (classical), and a quantum-kernel SVM using a `ZZFeatureMap` fidelity kernel $K(x_i, x_j) = |\langle \phi(x_i) | \phi(x_j) \rangle|^2$ computed exactly from statevectors. The dataset has 800 sessions of 6000 pulses each (attack ratio 0.47); results use 5-fold stratified cross-validation.

4 Results

All attacks are sub-threshold. Figure 1a shows the secure and attacked QBER distributions overlapping below the 11% line: only 0.8% of attack sessions exceed the threshold, so the operational test flags essentially none.

Headline: threshold vs. ML. Table 2 and Figure 2 contrast the operational QBER test with the random forest: 0.8% detection (AUC 0.726) versus 87.4% (AUC 0.963).

Table 2: Detection performance (5-fold cross-validated).

Detector	Detection rate @ 11%	ROC AUC
QBER threshold test	0.8 %	0.726
ML (rich features)	87.4 %	0.963

Per-attack analysis. Table 3 decomposes the result. The basis-biased attack raises QBER as a continuous score (AUC 0.931) but stays below the alarm level, so the operational threshold still misses it. The gain-matched PNS attack is the decisive case: QBER carries *no* signal (AUC 0.508,

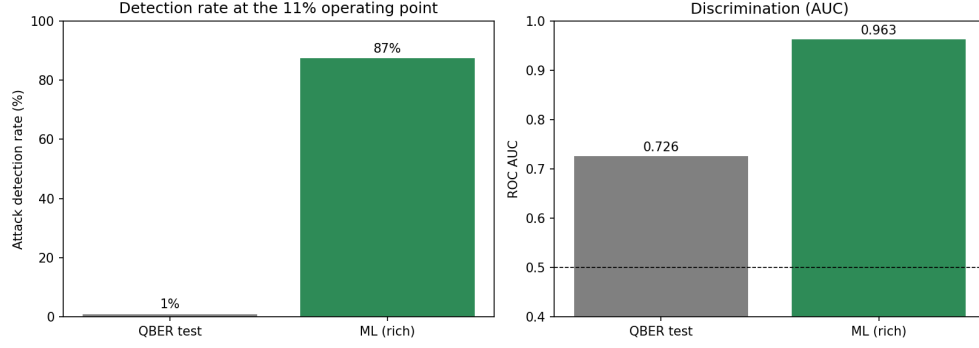


Figure 2: QBER threshold test (gray) versus ML on the full feature set (green): detection rate at the 11% operating point (left) and overall AUC (right).

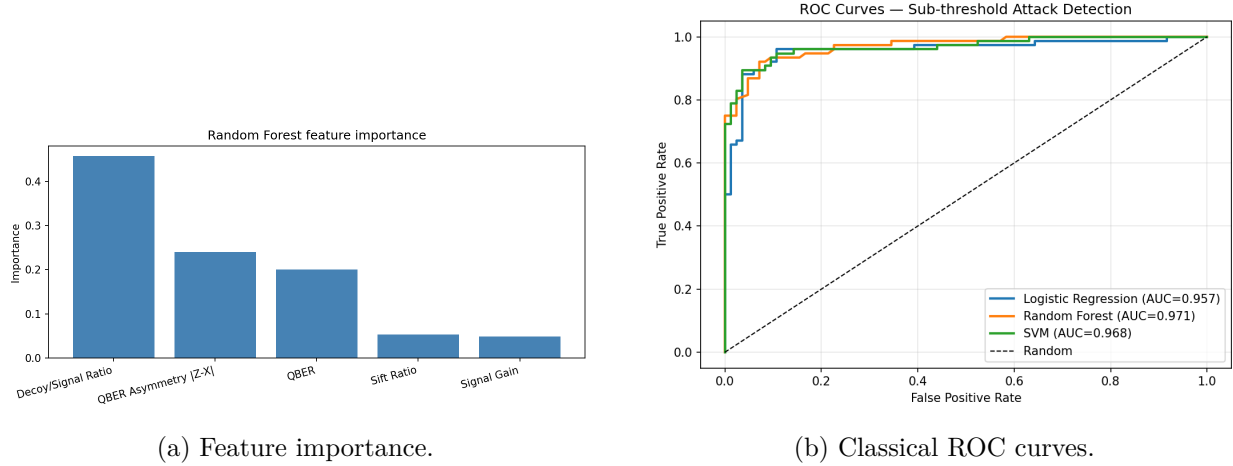


Figure 3: QBER is weak; the asymmetry and decoy ratio carry the signal.

random) and neither does the signal gain (AUC 0.57), yet ML detects it perfectly (AUC 1.000) through the decoy ratio alone.

Table 3: Secure-vs-attack discrimination, by attack family.

Attack	QBER-only AUC	ML AUC
Basis-biased intercept–resend	0.931	0.930
Gain-matched PNS	0.508	1.000

Feature importance and classical baselines. The random forest concentrates its importance on the QBER asymmetry and the decoy ratio, not QBER itself (Fig. 3a). The classical models are comparable (LogReg AUC 0.954 ± 0.012 , RF 0.963 ± 0.008 , SVM 0.956 ± 0.013); ROC curves in Figure 3b.

Classical vs. quantum, and kernel concentration. On an identical $n = 150$ training set the classical models reach AUC 0.94–0.965 (Fig. 4a). A naively-encoded quantum kernel ($[0, \pi]$ angles,

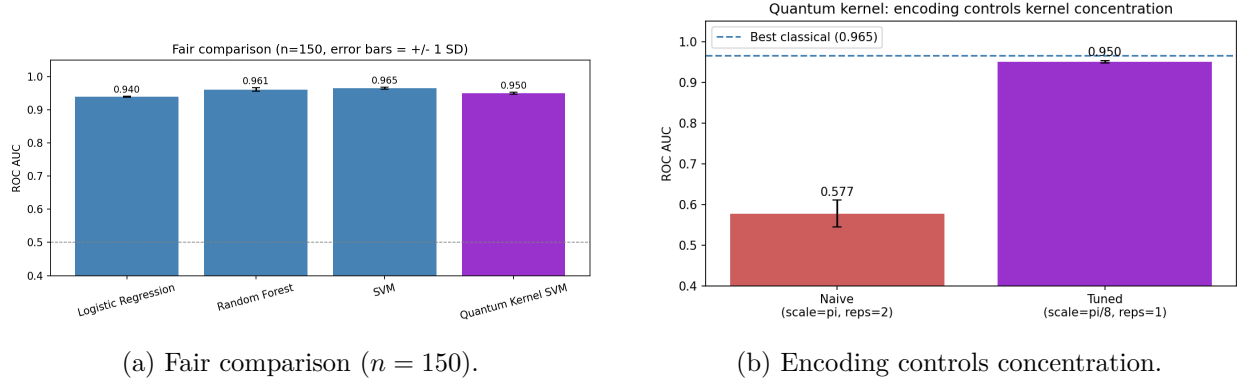


Figure 4: The quantum-kernel SVM matches classical only after the encoding is tuned to avoid exponential concentration ($0.58 \rightarrow 0.95$).

two repetitions) collapses to AUC 0.577: its kernel matrix is nearly the identity (off-diagonal mean 0.04) because the data states become near-orthogonal — *exponential kernel concentration*. Shrinking the encoding angles to $[0, \pi/8]$ with a single repetition restores AUC 0.950 (Fig. 4b), competitive with classical but offering no advantage.

5 Discussion

The central finding is operationally meaningful: the BB84 QBER threshold, used in practice as a go/no-go check, fails against attacks engineered to sit below it, whereas a lightweight classifier on already-available channel statistics recovers them. The attack mechanisms (basis bias, PNS) are known; the contribution is the demonstration that *cheap multivariate monitoring closes a detection gap the threshold leaves open*, with the per-basis asymmetry and decoy ratio identified as the informative observables. The gain-matched PNS result reproduces, through a data-driven lens, the exact reason decoy-state QKD exists.

We also report two honest negatives. A naive quantum kernel *fails* on this task due to exponential concentration; only after tuning does it match (never beat) classical methods — consistent with the literature on quantum kernels for low-dimensional classical data. We present this plainly rather than selecting a favourable configuration.

Limitations. The photon model is a simplified Monte-Carlo abstraction (no dark counts, after-pulsing, or finite-key corrections), the noise is idealised depolarizing, and only two attack families are studied. Extending to finite-key statistics, additional attacks (Trojan-horse, time-shift), and raw time-series features are natural next steps.

6 Conclusion

Sub-threshold eavesdropping defeats the standard BB84 QBER test but not machine learning on richer channel statistics: a random forest detects 87% of attacks that the threshold catches at 0.8%, and detects gain-matched PNS — invisible to both QBER and signal gain — perfectly via the decoy ratio. A quantum kernel offers no advantage and fails outright if naively configured. The work argues for treating QKD eavesdropping detection as a multivariate statistical problem rather than a single-threshold check.

References

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